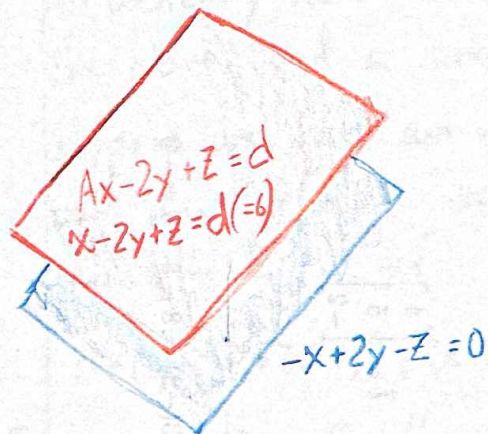


y Distance between planes.

If the distance between the plane $Ax - 2y + z = d$ and the plane containing the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is $\sqrt{6}$, then $|d|$ is?



Find some points on the blue plane. Let's evaluate line L_1 with equations $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ for a couple of values, starting with 0. To have them all equal 0, $x=1, y=2, z=3$ - point $(1, 2, 3)$ lies on the blue plane. Moving to another point, let's say we want the equations to equal 1, then $x=3, y=5, z=7$, point $(3, 5, 7)$ is also on the blue plane.

Let's evaluate the equations for line L_2 $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$, if we want these equations equal to zero we will get point $(2, 3, 4)$.

Now we have three points $(1, 2, 3)$, $(3, 5, 7)$ and $(2, 3, 4)$, let's use these points to figure out two vectors in the blue plane, that are not multiples of each other

$$A = (1, 2, 3) \quad \vec{a} = B - A = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$B = (3, 5, 7) \quad \vec{b} = C - A = \hat{i} + \hat{j} + \hat{k}$$

$$C = (2, 3, 4)$$

Now we can take the cross product of these to get a vector perpendicular to the blue plane:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 1 & 1 & 1 \end{vmatrix} = (3 \cdot 1 - 1 \cdot 4)\hat{i} - (2 \cdot 1 - 1 \cdot 4)\hat{j} + (2 \cdot 1 - 1 \cdot 3)\hat{k} \\ = -\hat{i} + 2\hat{j} - \hat{k} = \vec{n}$$

2) Distance between planes

Let's define, or find, the equation of the blue plane using the normal vector and a vector on the plane by taking their dot product and set it equal to zero.

An arbitrary vector on the plane can be defined as the difference between any arbitrary point on the plane, and a known point on the plane.

Suppose we have an arbitrary point (x, y, z) on the plane, and using our known point $(3, 5, 7)$, we can now define a vector on the plane as $(x-3)\hat{i} + (y-5)\hat{j} + (z-7)\hat{k}$, and taking the dot product of this with $\vec{n}$, we get

$$\vec{n} \cdot (x-3)\hat{i} + (y-5)\hat{j} + (z-7)\hat{k} = (-1)(x-3) + 2(y-5) - 1(z-7) = -x+3+2y-10-z+7 = -x+2y-z=0$$

The equation for the blue plane therefore is: $\boxed{-x+2y-z=0}$

In order to talk about the distance between two planes, they have to be parallel. That means the ratio of coefficient A, B, C and D between the planes has got to be the same.

Inspecting the equations for both planes:

red plane: $Ax - 2y + z = d$
blue plane: $-x + 2y - z = 0$

We notice that by multiplying the blue plane's equation with (-1) we get the value of A , and we see the ratio of the coefficients are the same.

$$-x + 2y - z = 0 \cdot (-1) = x - 2y + z \Rightarrow A = 1$$

Now we can figure out the distance between these two planes

3) distance between planes

We can use the formula for point distance to plane: $\frac{Ax_0 + By_0 + Cz_0 - D}{\sqrt{A^2 + B^2 + C^2}}$

By evaluating the plane equation using a known point over the magnitude of $\vec{n}$ using the point $A=(1,2,3)$, evaluated in the equation for the red plane $x-2y+z=d$, we can write:

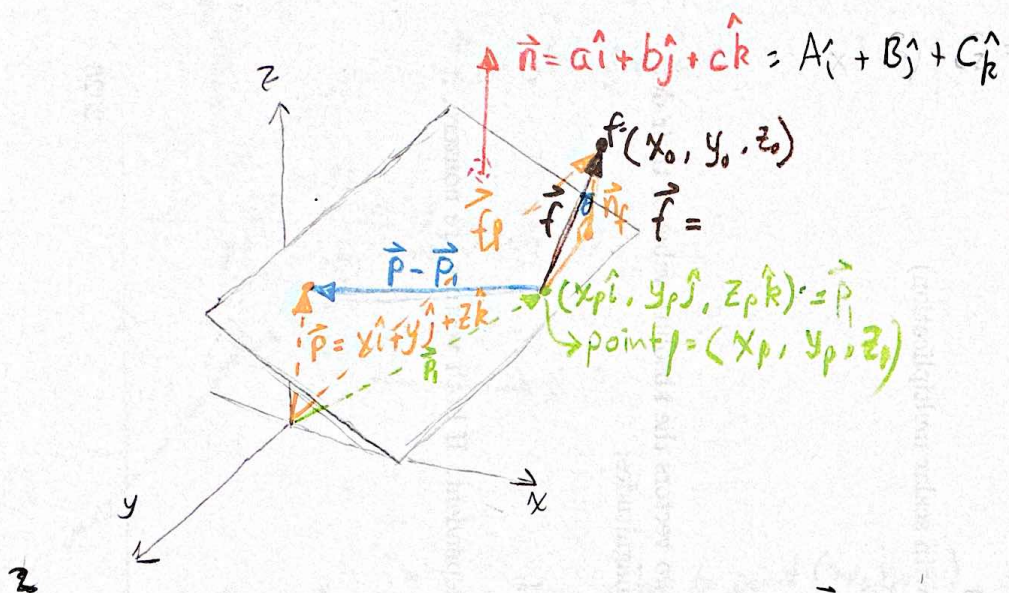
$$\text{distance} = \frac{1 - 4 + 3 - d}{\sqrt{1 + 4 + 1}} = \frac{-d}{\sqrt{6}} \Rightarrow \text{They give us the distance in the assignment, } = \sqrt{6}$$

substituting distance with $\sqrt{6}$, we get

$$\sqrt{6} = \frac{-d}{\sqrt{6}} \Rightarrow d = -6 \quad |d| = 6$$

This means the equation for the red plane = $x - 2y + z = 6$

Normal vector from plane equation



Introducing θ , trigonometry gives us $\cos \theta = \frac{\vec{n}_f}{\vec{f}}$, which equals $\vec{n}_f = \vec{f} \cos \theta$.

Given we don't know the value of θ , we can use the dot product. $\vec{n}_f \cdot \vec{f} = \|\vec{n}_f\| \|\vec{f}\| \cos \theta$
By performing this little trick:

$$\vec{n}_f = \|\vec{f}\| \cos \theta \cdot \frac{\|\vec{n}\|}{\|\vec{n}\|} \quad (\text{we know the normal vector})$$

$$\vec{n}_f = \frac{\|\vec{n}\| \cdot \|\vec{f}\| \cos \theta}{\|\vec{n}\|} \quad \text{this now equals the dot product of } \vec{n} \cdot \vec{f}$$

$$\frac{\vec{n} \cdot \vec{f}}{\|\vec{n}\|} = \frac{Ax_0 - Ax_p + By_0 - By_p + Cz_0 - Cz_p}{\sqrt{A^2 + B^2 + C^2}} = \|\vec{n}_f\| =$$

Point distance to plane

Say we have a plane defined by two position vectors $\vec{p}$ and $\vec{p}_1$.

Suppose we want to know the distance from point f , defined by the position vector $\vec{f}$ to the plane. By distance we mean the shortest distance.

Let's start by constructing a vector from a point in the plane, p , to the point outside the plane, f . This is the vector $\vec{f}$.

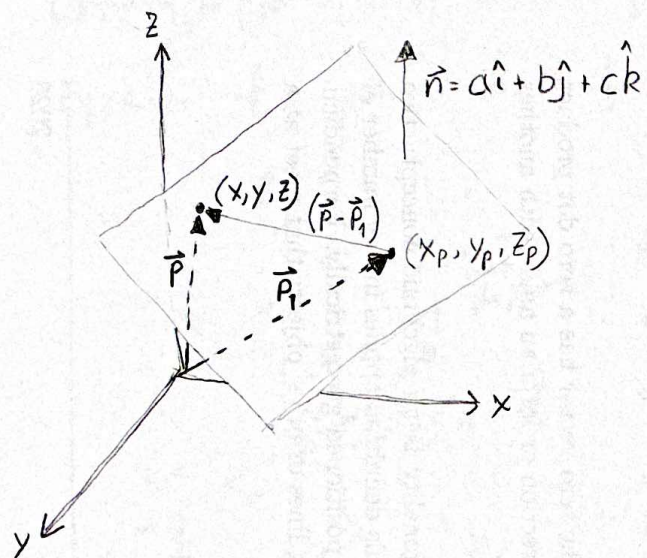
$$\vec{f} = \vec{f}_p - \vec{p}_1$$

$$\vec{f} = (x_0 - x_p)\hat{i} + (y_0 - y_p)\hat{j} + (z_0 - z_p)\hat{k}$$

The shortest distance from f to the plane will be its normal vector $\vec{n}_f$.

$$\frac{Ax_0 + By_0 + Cz_0 - D}{\sqrt{A^2 + B^2 + C^2}}$$

Normal vector from plane



$\vec{P}_1 = x_p \hat{i} + y_p \hat{j} + z_p \hat{k}$ - position vector to point (x_p, y_p, z_p)

$\vec{P} = x \hat{i} + y \hat{j} + z \hat{k}$ - position vector to point (x, y, z)

Both points are within the plane

$(\vec{P} - \vec{P}_1)$ - vector that lies within the plane

$\vec{n} = a \hat{i} + b \hat{j} + c \hat{k}$ Normal vector to the plane

With this setup we can use the dot product to find a normal vector to the plane simply by setting the dot product to be zero.

$\vec{n} \cdot (\vec{P} - \vec{P}_1) = 0 \Rightarrow$ this will give us the equation for $\vec{n}$

$$\vec{P} - \vec{P}_1 = (x - x_p) \hat{i} + (y - y_p) \hat{j} + (z - z_p) \hat{k}$$

$$\vec{n} \cdot (\vec{P} - \vec{P}_1) = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} x - x_p \\ y - y_p \\ z - z_p \end{bmatrix} = ax - ax_p + by - by_p + cz - cz_p$$

∴ Now setting the dot product to zero

$$ax - ax_p + by - by_p + cz - cz_p = 0$$

Rewrite:

$$ax + by + cz = ax_p + by_p + cz_p$$

Given the equation for a plane:

$$Ax + By + Cz = D \rightarrow \text{move the plane along } x \text{ axis}$$

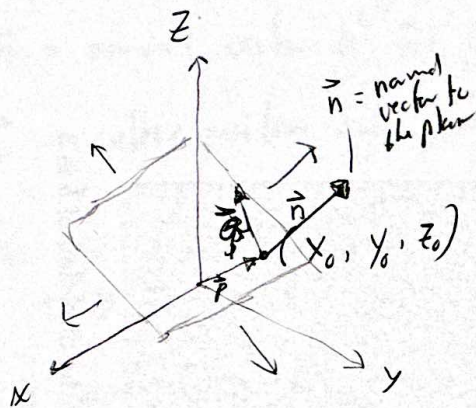
$$ax + by + cz = \overbrace{ax_p + by_p + cz_p}$$

We see now that given the equation for a plane, we can easily find the equation for a normal vector to that plane

For any plane $Ax + By + Cz = D$, the normal vector $\vec{n} = A \hat{i} + B \hat{j} + C \hat{k}$

Defining a plane in $\mathbb{R}^3$ with a point and a normal vector

Equation of a plane $Ax + By + Cz = D$



Another way to define a plane is to give you a point on the plane, and a vector that is perpendicular to the plane

The point can be specified by a position vector $\vec{p}$

$$\vec{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\vec{n} \cdot \vec{r} = 0$$

~~$\vec{p}$ = vector in the plane~~
 $\vec{n}$ = vector normal to the plane
 $\vec{p}$ = position vector to point
 $\vec{r}$ = vector in the plane

Defining a point on a plane in $\mathbb{R}^3$ with a point and a normal vector

$\vec{x}_0$ - position vector: $\begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$, pointing to the point (x_0, y_0, z_0) which lies on the plane
 $\vec{n}$ - normal vector to the plane: $\begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$
 $\vec{x}$ - another position vector to the plane: $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

This means that $(\vec{x} - \vec{x}_0)$ must lie in the plane, and be orthogonal to $\vec{n}$.

This means if we set the dot product $\vec{n} \cdot (\vec{x} - \vec{x}_0)$ to zero, they are orthogonal

$$\vec{n} \cdot (\vec{x} - \vec{x}_0) = 0$$

$$\begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} \cdot \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix} = 0$$

$$n_1(x - x_0) + n_2(y - y_0) + n_3(z - z_0) = 0 \Rightarrow \text{This is the same as the equation for a plane}$$

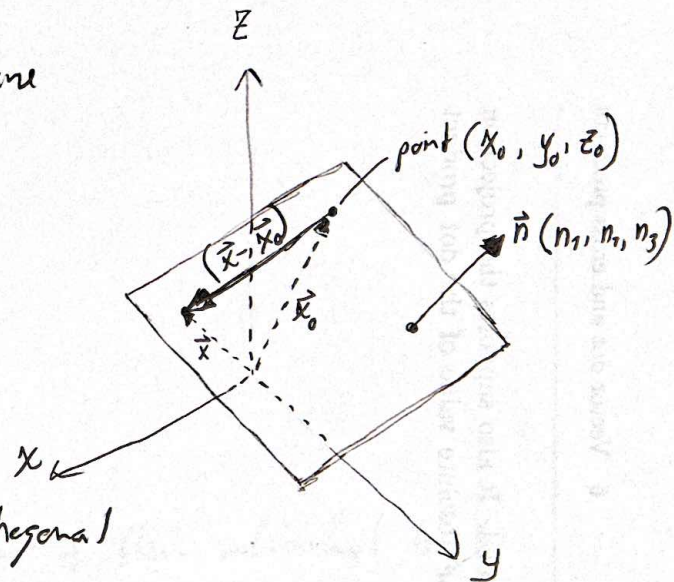
$$\boxed{Ax + By + Cz = D}$$

Suppose we have

$$\vec{n} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, \quad \vec{x}_0 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \Rightarrow 1(x-1) + 3(y-2) - 2(z-3) = 0$$

$$xA + 3y - 2z = 1$$

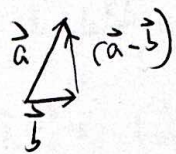
$$\boxed{x + 3y - 2z = 1}$$



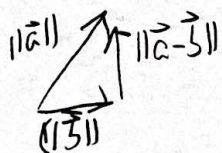
Defining the angle between two vectors

$\vec{a}, \vec{b} \in \mathbb{R}^n$, nonzero vectors

1)



2) ^{constructed} another triangle when the sides are defined by vector lengths



As for any triangle:

$$\left. \begin{aligned} ||\vec{b}|| &> ||\vec{a}|| + ||\vec{a}-\vec{b}|| \\ ||\vec{a}|| &> ||\vec{a}-\vec{b}|| + ||\vec{b}|| \\ ||\vec{a}-\vec{b}|| &> ||\vec{a}|| + ||\vec{b}|| \end{aligned} \right\} \begin{array}{l} \text{we cannot} \\ \text{construct triangle} \end{array}$$

3) vector triangle inequality

The sum of two vectors, is always going to be less than or equal to the sum of their individual lengths

$$||\vec{x} + \vec{y}|| \geq ||\vec{x}|| + ||\vec{y}||$$

4) Law of cosines $c^2 = a^2 + b^2 - 2ab \cos \theta$

Applied to vector triangle:

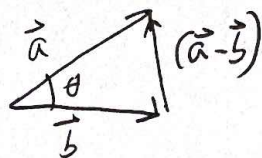
$$||\vec{a} - \vec{b}||^2 = ||\vec{a}||^2 + ||\vec{b}||^2 - 2||\vec{a}|| ||\vec{b}|| \cos \theta$$

expanding $||\vec{a} - \vec{b}||^2$

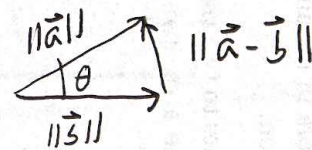
$$||\vec{a} - \vec{b}||^2 = ||\vec{a} - \vec{b}|| \cdot ||\vec{a} - \vec{b}||$$

$$= ||\vec{a}||^2 - ||\vec{a}|| ||\vec{b}|| - ||\vec{b}|| ||\vec{a}|| + ||\vec{b}||^2$$

$$\boxed{= ||\vec{a}||^2 - 2||\vec{a}|| ||\vec{b}|| + ||\vec{b}||^2}$$



$\Rightarrow$

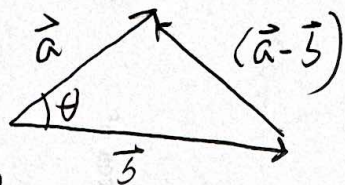


$$||\vec{a}||^2 - 2||\vec{a}|| ||\vec{b}|| + ||\vec{b}||^2 = ||\vec{a}||^2 + ||\vec{b}||^2 - 2||\vec{a}|| ||\vec{b}|| \cos \theta$$

simplify

$$||\vec{a}|| ||\vec{b}|| =$$

Law of cosines applied to vectors



$$\|\vec{a} - \vec{b}\|^2 = \|\vec{a}\|^2 + \|\vec{b}\|^2 - 2\|\vec{a}\|\|\vec{b}\|\cos\theta$$

expand the left side:

$$\|\vec{a} - \vec{b}\|^2 = (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b}) = \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} = \boxed{\|\vec{a}\|^2 - 2(\vec{a} \cdot \vec{b}) + \|\vec{b}\|^2}$$

insert expanded left side to law of cosines:

$$\|\vec{a}\|^2 - 2(\vec{a} \cdot \vec{b}) + \|\vec{b}\|^2 = \|\vec{a}\|^2 + \|\vec{b}\|^2 - 2\|\vec{a}\|\|\vec{b}\|\cos\theta$$

we can simplify:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos\theta \rightarrow \cos\theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|\|\vec{b}\|} \Rightarrow$$

Angle between two vectors in $\mathbb{R}^n$

$$\theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|\|\vec{b}\|} \right)$$

$$\vec{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \vec{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 90^\circ$$

$$\vec{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \vec{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \cos^{-1} \frac{\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}}{\sqrt{1^2+0^2} \sqrt{1^2+1^2}} = \frac{1}{\sqrt{2}} = \cos^{-1} \left(\frac{1}{\sqrt{2}} \right) = \frac{\pi}{4} = 45^\circ$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1 \cdot 1) + (0 \cdot 1) = 1$$

$$\sqrt{1^2+0^2} \cdot \sqrt{1^2+1^2} = \sqrt{2}$$

